

Statistical Questions and Describing Variability

Answer Key

Grade 6

Quick Answers

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| 1. — | 7. (b) the range of the ages = 37, — |
| 2. 18 | 8. (a) range for the Owls = 4, (b) range for the Robins = 24, — |
| 3. (b) the median number of pets = 1, — | 9. (a) busiest day = Thu, (b) total for the week = 28, (c) the range of the daily counts = 6 |
| 4. (a) lowest temperature = 11, (b) highest temperature = 22, (c) the range = 11 | 10. (a) the mean wait = 8, (b) the median wait = 4, (c) the range = 44, — |
| 5. (a) the mean = 8, (b) the median = 8, — | |
| 6. (a) the lower quartile = 6, (b) the upper quartile = 16, (c) the interquartile range = 10 | |
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What is verified

4 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 3, 5, 7, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6

6.SP.A – *problems 1, 3, 5, 7, 8, 10*

6.SP.B.5 – *problems 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 25 tagged checks.

1. Which of the two piano-club questions is statistical?

Solution: Instructor-judged. Question 2 is the statistical one. Asking how many minutes the members of the club practise each day is answered by collecting many values, and those values differ from member to member — that variability is the test. Question 1 names one student and one day, so it has a single answer and nothing varies. Full credit needs the choice and a reason that mentions the answers varying.

2. Range of the reading times 12, 20, 15, 30, 18, 25, 22.

Solution: The largest value is 30 minutes and the smallest is 12, so the range is $30 - 12 = 18$ minutes.

18

3. Pets owned by eight students: 0, 1, 2, 1, 3, 0, 2, 1.

Solution: (a) **Instructor-judged.** Any question about the whole group whose answers vary earns credit — for example “how many pets does a student in this group typically own?” or “how spread out are the numbers of pets?”. A question with one fixed answer, such as how many pets the group owns altogether, does not: nothing varies in it.

(b) In order the values are 0, 0, 1, 1, 1, 2, 2, 3. With eight values the median is the mean of the fourth and fifth: $(1 + 1) \div 2 = 1$ pet. 1

4. Six daily high temperatures: 14, 19, 11, 22, 17, 20.

Solution: (a) Reading down the row, the lowest is 11 degrees. 11

(b) The highest is 22 degrees. 22

(c) Range = $22 - 11 = 11$ degrees. Note that the range is a single number, not a pair — it measures how far the values reach, not where they are. 11

5. Ten spelling marks: 7, 8, 9, 8, 6, 10, 8, 7, 9, 8.

Solution: (a) The marks total 80, and $80 \div 10 = 8$. 8

(b) In order: 6, 7, 7, 8, 8, 8, 8, 9, 9, 10. The fifth and sixth are both 8, so the median is 8. 8

(c) **Instructor-judged.** A correct description gives a centre *and* a spread: the marks cluster at 8, where the mean and the median agree, and every value lies between 6 and 10, so the set is tightly grouped with nothing far from the middle. Full credit needs the typical value and a statement about how spread out the values are.

6. Twelve plant heights in order: 2, 4, 5, 7, 8, 10, 11, 13, 15, 17, 19, 21.

Solution: (a) The lower half is 2, 4, 5, 7, 8, 10, and its median is $(5+7) \div 2 = 6$ centimetres. 6

(b) The upper half is 11, 13, 15, 17, 19, 21, and its median is $(15+17) \div 2 = 16$ centimetres. 16

(c) Interquartile range = $16 - 6 = 10$ centimetres. This is the width of the middle half of the data, which is why one very tall plant would not change it. 10

7. Ages in one family: 4, 9, 12, 38, 41.

Solution: (a) **Instructor-judged.** No — as it stands it is not a statistical question. Asking how old each person in this one family is has a fixed list of answers; there is nothing to collect and nothing that varies from one asking to the next. Full credit also goes to a response that rewrites it as a statistical question, such as asking about the ages in many families. Answering “no” with no reason does not earn credit.

(b) Range = $41 - 4 = 37$ years. 37

8. Owls 20, 21, 22, 23, 24 and Robins 10, 16, 22, 28, 34; both mean 22.

Solution: (a) Owls: $24 - 20 = 4$ points.

4

(b) Robins: $34 - 10 = 24$ points.

24

(c) **Instructor-judged.** The Owls are the more consistent team. Both teams average 22 points, so the mean alone says the two are identical; the ranges say otherwise — the Owls' scores lie inside a 4-point band while the Robins' spread across 24 points. Full credit needs the comparison of the two ranges, not just the observation that the means are equal.

9. Library borrowers: Mon 4, Tue 7, Wed 3, Thu 9, Fri 5.

Solution: (a) The largest count is 9, on Thursday.

Thu

(b) $4 + 7 + 3 + 9 + 5 = 28$ borrowers in the week.

28

(c) The counts run from 3 up to 9, so the range is $9 - 3 = 6$ borrowers.

6

10. Nine waiting times: 1, 2, 2, 3, 4, 4, 5, 6, 45 minutes.

Solution: (a) The times total 72 minutes, and $72 \div 9 = 8$ minutes.

8

(b) The values are already in order and there are nine of them, so the median is the fifth: 4 minutes.

4

(c) Range = $45 - 1 = 44$ minutes.

44

(d) **Instructor-judged.** The median describes a typical wait better. One person waited 45 minutes, and that single value pulls the mean up to 8 minutes — longer than eight of the nine waits actually recorded — while leaving the median at 4 minutes, where the bulk of the data sits. Full credit needs the effect of the one extreme value on the mean and a reasoned choice between the two measures.
